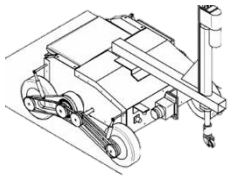
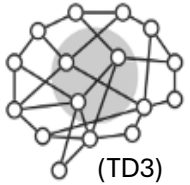
Curriculum

Seguimiento y Curriculum



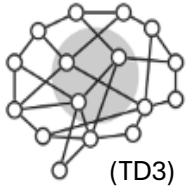
Agente RL untrained



Entrenamiento:
Scen32, Reward_SystemF2
Run_x316, 1100eps

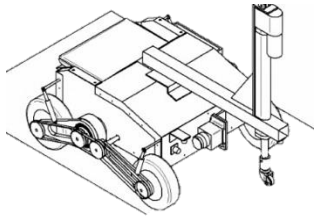
Agente 100

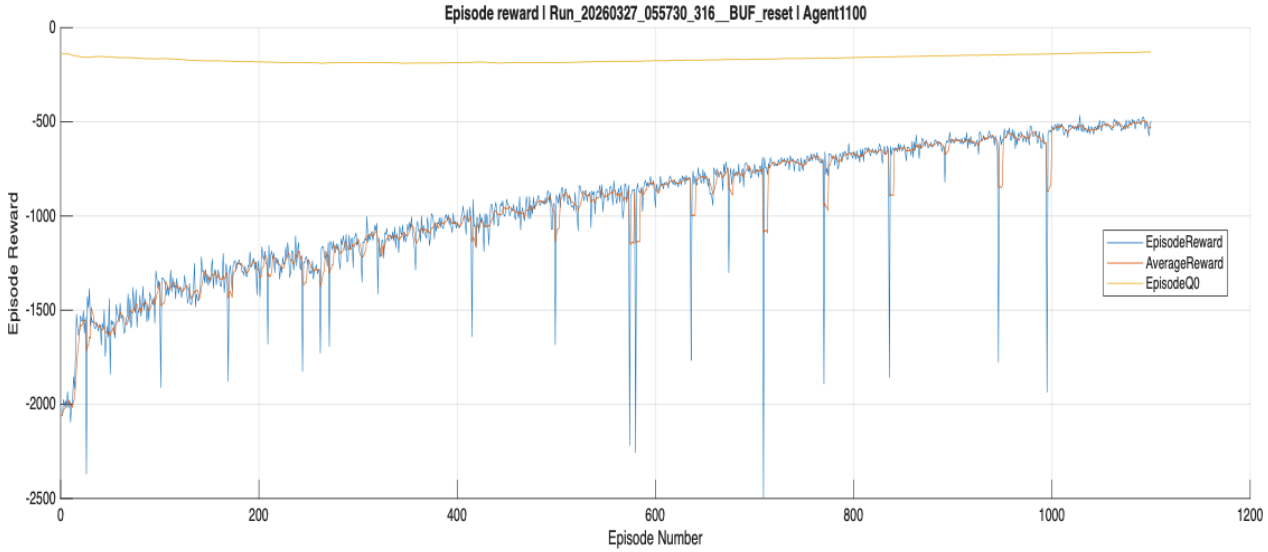
Agente seleccionado
Checkpoint100



Evaluación del Aprendizaje

Curva de Aprendizaje y Métricas. Agente1100



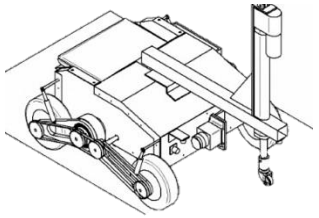


La Fase 2 con recompensa basada principalmente en el error de seguimiento evidencia una convergencia progresiva del retorno acumulado, con mejora sostenida del desempeño y estabilidad creciente del agente a lo largo del entrenamiento.

Training Stats			**Advanced Agent Options**		
Métrica	Símbolo	Valor	Métrica	Símbolo	Valor
Total de episodios	N_{ep}	1100	Tiempo muestreo	T_s	0.01
Recompensa total último episodio	R_{ep}	-495.652	Factor descuento	γ	0.99
Recompensa Media	$\bar{R}$	-528.005	Tamaño del Batchize	B	64
Pasos último episodio	-	600	Tamaño del Exp. Buffer		10000
Número total de pasos	N_{steps}	660000	Modo del Buffer	-	Reset
Promedio pasos/eps	-	600	Taza aprendizaje actor	α_a	1.0000e-04
Valor Q estimado	Q_0	-129.333	Optimizador del actor	-	adam
			Taza aprendizaje critico	α_c	7.0000e-04
			Optimizador del critico	-	adam
			Coeficiente actualización suave	τ	0.005

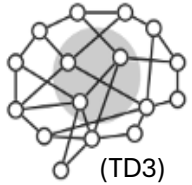
Exploration Gaussian Model			**Target Policy Smoothing**		
Métrica	Símbolo	0.3162	Métrica	Símbolo	Valor
Std.Dev	σ	0	Std.Dev	σ	0.2
Mean	μ	-0.5	Mean	μ	0
Lower Lim	-	0.5	Lower Lim	-	-0.5
Upper Lim	-	1.7442e-06	Upper Lim	-	0.5
Std.Dev decay	λ_σ	0.1	Std.Dev decay	λ_σ	4.9929e-06
Std.Dev min	σ_{min}	0.3162	Std.Dev min	σ_{min}	0.01

Evaluación del Aprendizaje

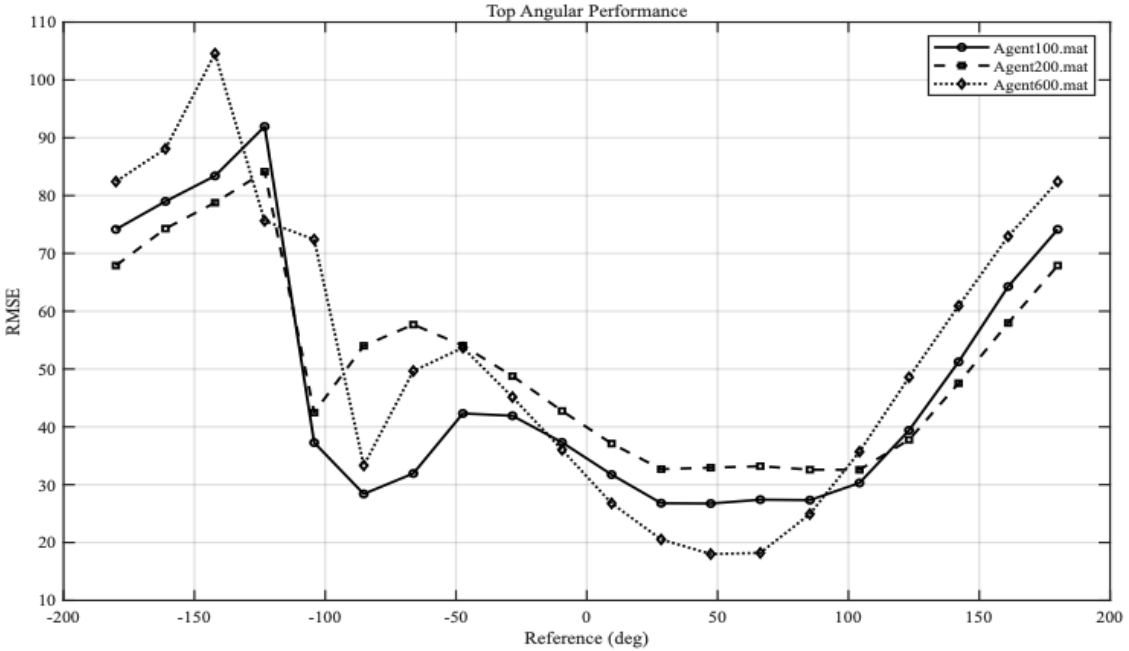


Top3 Curvas RMSE en el Dominio Angular

Agentes RL



(TD3)

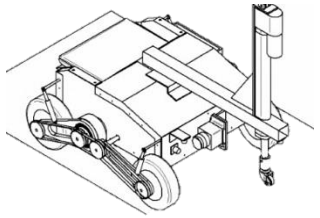


La figura muestra el RMSE global de los tres mejores agentes en todo el dominio angular. Se observa un incremento del error hacia los extremos (±180°); sin embargo, no existe simetría en la distribución del RMSE, lo que indica un desbalance en el desempeño del agente según la región angular evaluada.

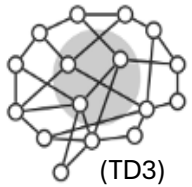
Evaluación del Aprendizaje

Universidad Internacional de Valencia

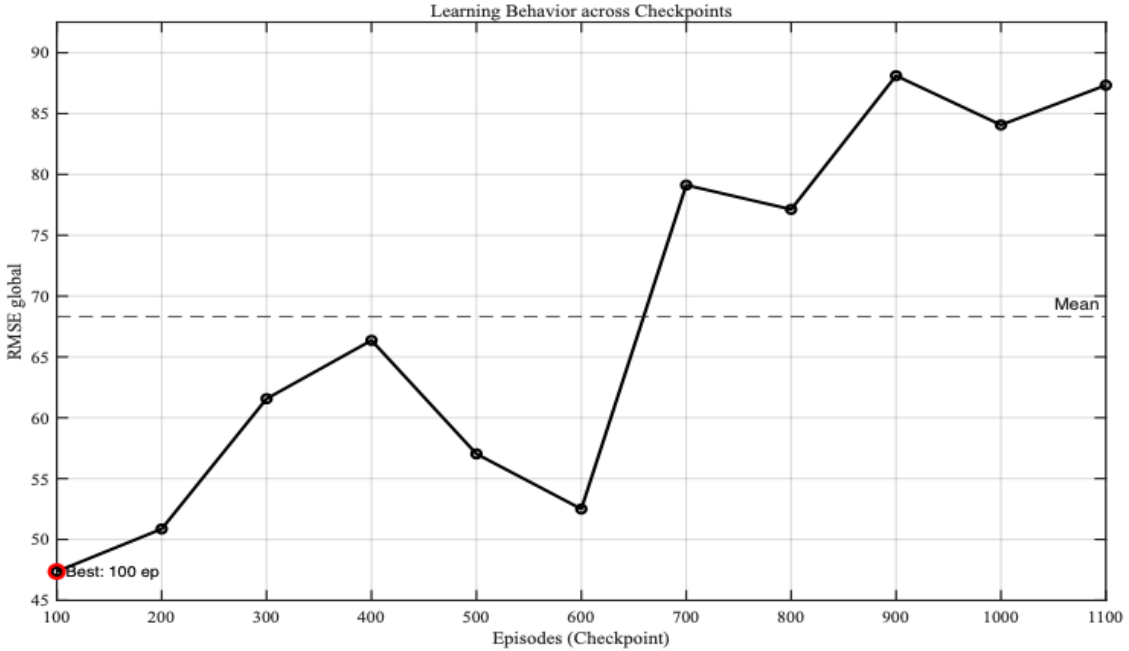
Progresión del RMSE durante el entrenamiento



Agentes RL



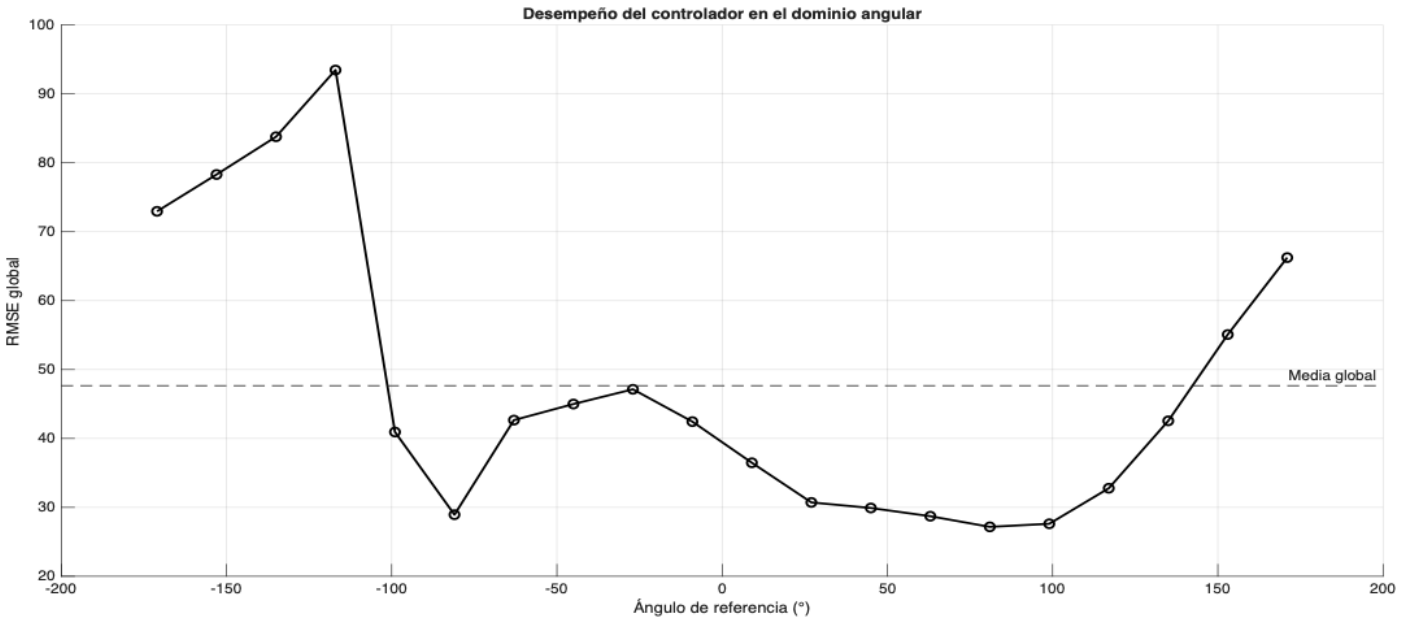
(TD3)

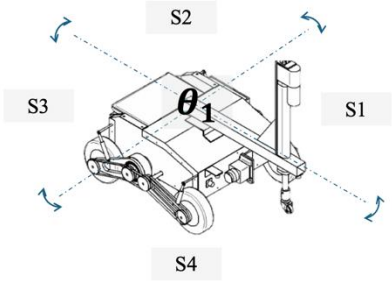


La evolución del RMSE demuestra que extender el entrenamiento no conduce a mejoras sostenidas del desempeño, evidenciando un punto de saturación en la capacidad de generalización del agente

Evaluación en el Dominio Angular en Simulación

$RMSE_i$ en dominio angular





Dominio Angular: [-180°, 180°]

$180° \leq \theta_1 \leq 90°$	$90° < \theta_1 \leq 0°$	$0° > \theta_1 \geq -90°$	$-90° > \theta_1 \geq -180°$
S2	S1	S4	S3

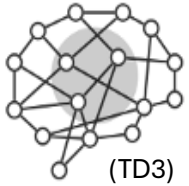
el Agente100 mantiene un desempeño general en la región central, pero presenta una degradación progresiva hacia los extremos, con un comportamiento no simétrico entre sectores. Se observa una pérdida abrupta de precisión en determinados rangos angulares, lo que sugiere limitaciones en la cobertura del entrenamiento y una generalización parcial de la política aprendida.

El agente generaliza, pero pierde precisión

Evaluación del Aprendizaje

Selección del Agente en base a su rendimiento. Evaluación Determinista en la simulación

Entrenamiento:
Scen32, Reward_SystemF2
Run_x316, 1100eps



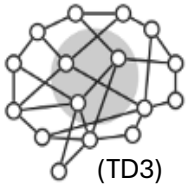
Agente seleccionado
Checkpoint100

Tabla con posición de los agentes según sus métricas

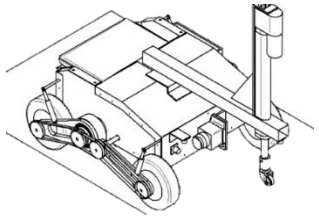
Position	Agente	$RMSE_{global}$	$RMSE_{max}$	$\sigma_{angular}$	$Ratio_{ext}$	AUC
1	Agent100.mat	47.3593	91.9526	21.9152	1.9416	45.9488
2	Agent200.mat	50.8623	84.1346	16.6631	1.6542	49.9653
3	Agent600.mat	52.5019	104.5536	26.1521	1.9914	50.9272
4	Agent500.mat	57.0328	104.2105	22.3312	1.8272	56.4216
5	Agent300.mat	61.5651	139.2792	32.2617	2.2623	60.6632
6	Agent400.mat	66.3630	137.8620	33.3298	2.0774	65.1121
7	Agent800.mat	77.1289	137.6104	46.5498	1.7842	74.0597
8	Agent700.mat	79.1222	150.7403	48.0584	1.9052	75.3528
9	Agent1000.mat	84.0683	142.5539	45.6387	1.6957	81.6264
10	Agent1100.mat	87.3358	165.9936	49.7643	1.9006	84.6944

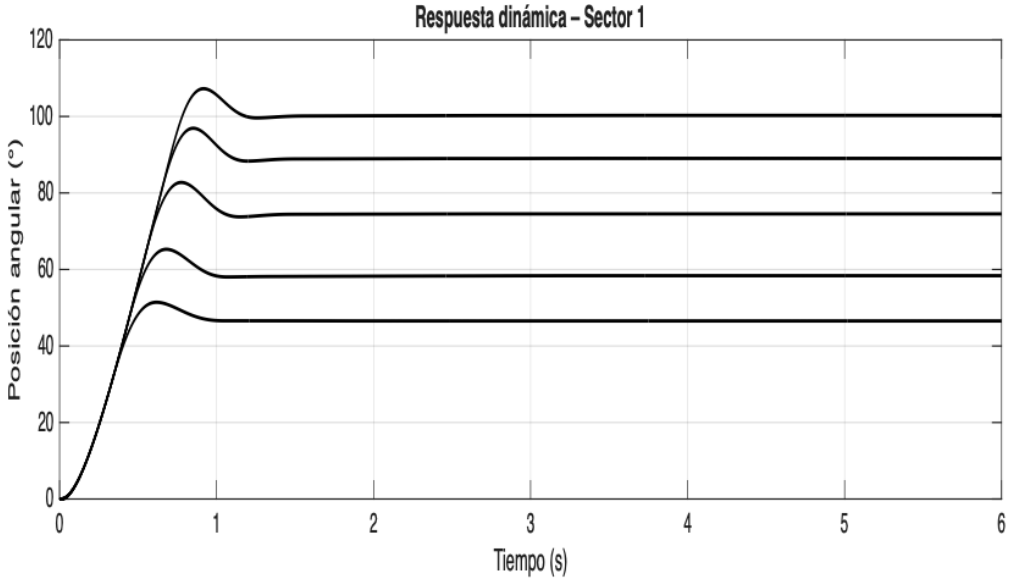
La evaluación determinista permite comparar objetivamente su desempeño en el dominio angular. En este escenario, el **Agent100** se posiciona como el mejor candidato, al minimizar el RMSE global y mantener un equilibrio favorable entre error máximo, dispersión angular y rendimiento global

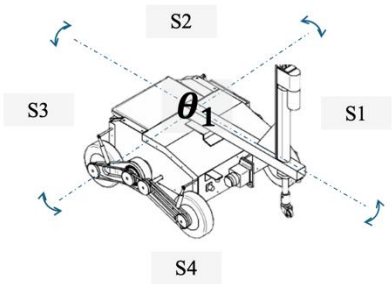
Evaluación Determinista en la Simulación



Respuesta dinámica del sector 1







Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

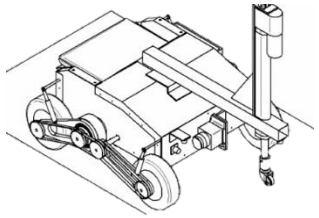
Rank	O1sp	O1_final	Mp [°]	Mp[%]	T[s]	e_ss_abs[°]	RMSE[°]	ITAE[°s]	Tr[s]	Iosc	U_peak	U_rms	U_energy	U_abs	U_TV
1	81.00	100.28	26.26	32.42	6.00	19.28	27.17	350.20	0.45	0.50	9.79	3.63	7912.13	1502.81	81.00
2	63.00	89.04	33.92	53.84	6.00	26.04	28.72	468.74	0.37	0.47	9.83	3.74	8402.64	1683.86	63.00
3	45.00	74.53	37.75	83.89	6.00	29.53	29.91	530.59	0.28	0.43	9.88	3.75	8451.52	1824.92	45.00
4	27.00	58.38	38.28	141.78	6.00	31.38	30.72	563.47	0.20	0.36	9.90	3.65	8000.41	1880.43	27.00
5	9.00	46.55	42.43	471.44	6.00	37.55	36.48	674.88	0.10	0.30	9.85	3.49	7327.97	1900.78	9.00

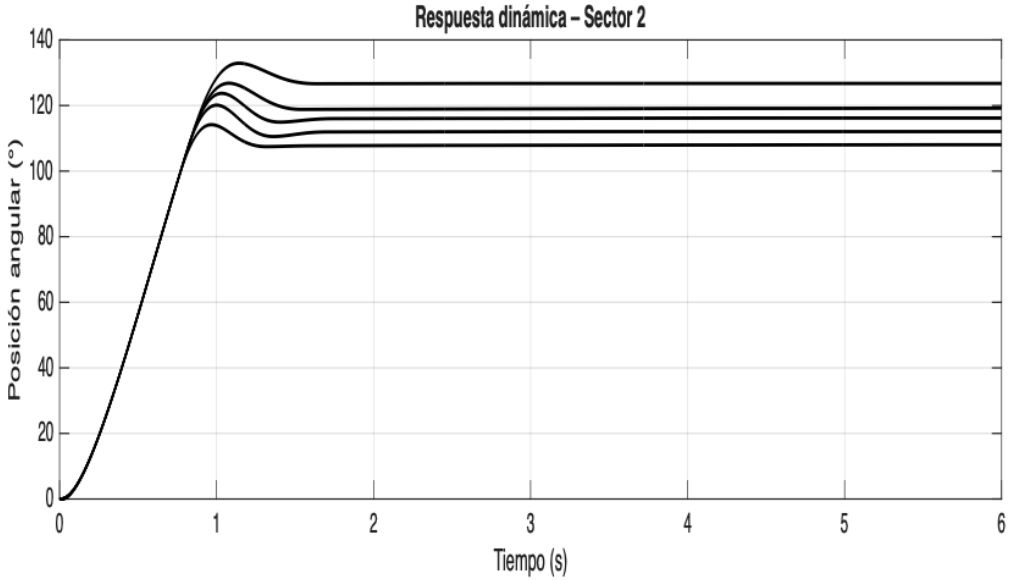
Evaluación Determinista en la Simulación

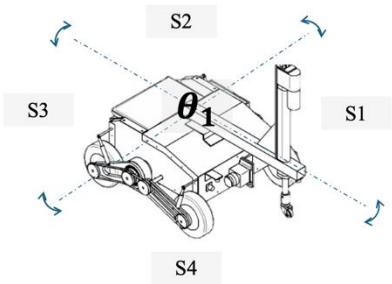
Universidad Internacional de Valencia

(TD3)

Respuesta dinámica del sector 2



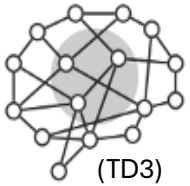




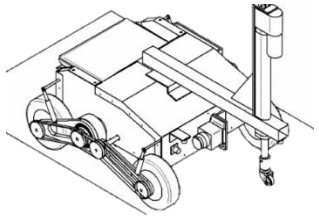
Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

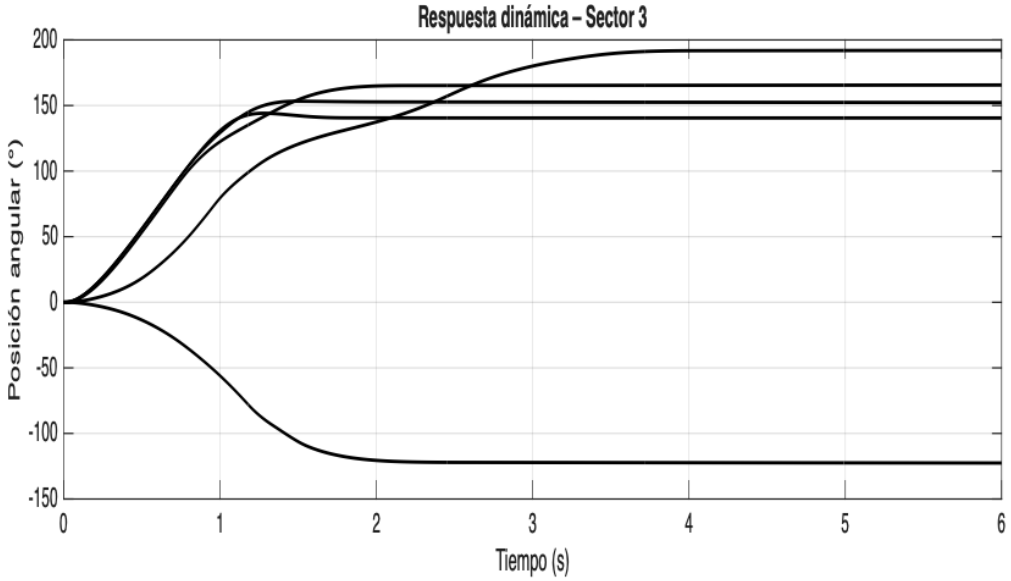
Rank	O1sp	O1_final	Mp [°]	Mp[%]	T[s]	e_ss_abs[°]	RMSE[°]	ITAE[°s]	Tr[s]	Iosc	U_peak	U_rms	U_energy	U_abs	U_TV
1	99.00	108.04	15.15	15.31	6.00	9.04	27.62	171.15	0.53	0.51	9.80	3.52	7441.38	1359.65	40.96
2	117.00	112.05	3.12	2.67	6.00	4.95	32.78	104.66	0.62	0.53	9.85	3.53	7503.54	1330.74	41.32
3	135.00	116.19	-11.29	-8.36	6.00	18.81	42.56	355.33	0.75	0.54	9.89	3.45	7141.05	1221.30	39.79
4	153.00	119.19	-26.20	-17.12	6.00	33.82	55.08	626.61	NaN	0.54	9.94	3.33	6654.51	1090.77	36.18
5	171.00	126.73	-38.10	-22.28	6.00	44.27	66.25	816.64	NaN	0.55	9.86	3.17	6022.88	937.60	33.67

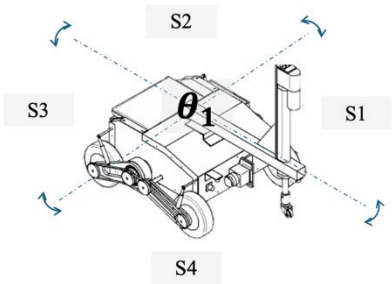
Evaluación Determinista en la Simulación



Respuesta dinámica del sector 3



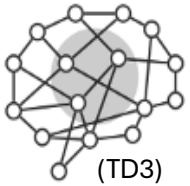




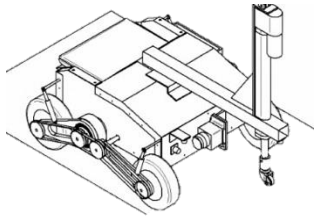
Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

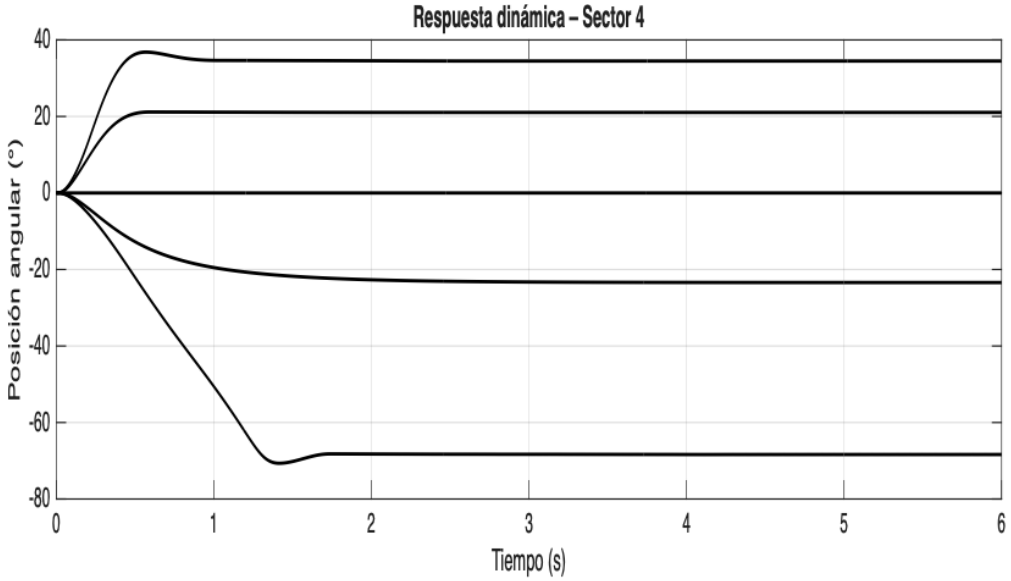
Rank	O1sp	O1_final	Mp [°]	Mp[%]	T[s]	e_ss_abs[°]	RMSE[°]	ITAE[°s]	Tr[s]	Iosc	U_peak	U_rms	U_energy	U_abs	U_TV
1	-99.00	-122.54	99.00	100.00	6.00	23.51	40.94	432.79	0.00	0.41	4.45	2.59	4033.51	1530.41	20.38
2	-171.00	140.51	315.09	184.27	6.00	48.49	72.97	900.33	0.00	0.57	9.46	2.77	4598.08	694.69	32.26
3	-153.00	152.27	306.24	200.16	6.00	54.70	78.31	1018.61	0.00	0.57	8.26	2.15	2763.53	727.36	33.52
4	-135.00	165.63	300.63	222.69	6.00	59.40	83.79	1129.98	0.00	0.56	5.44	1.73	1797.48	951.42	34.51
5	-117.00	192.04	309.04	264.14	6.00	51.00	93.47	1168.31	0.00	0.50	5.21	2.04	2485.86	1082.85	27.97

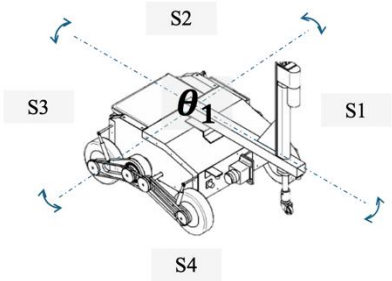
Evaluación Determinista en la Simulación



Respuesta dinámica del sector 4







Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

Rank	O1sp	O1_final	Mp [°]	Mp[%]	T[s]	e_ss_abs[°]	RMSE[°]	ITAE[°s]	Tr[s]	Iosc	U_peak	U_rms	U_energy	U_abs	U_TV
1	-81.00	-68.37	81.00	100.00	6.00	12.63	28.94	248.20	0.00	0.25	5.69	1.55	1438.80	778.16	19.71
2	-9.00	34.49	45.82	509.13	6.00	43.49	42.44	782.27	0.00	0.23	9.08	3.21	6179.72	1844.20	33.97
3	-63.00	-23.41	63.00	100.00	6.00	39.59	42.67	720.50	0.00	0.09	4.99	1.51	1356.05	855.23	5.44
4	-45.00	0.00	45.00	100.00	6.00	45.00	45.00	810.00	0.00	0.00	3.24	0.78	364.62	465.78	2.47
5	-27.00	21.05	48.16	178.37	6.00	48.05	47.14	864.20	0.00	0.13	5.21	2.70	4369.58	1604.21	22.89